

The Section-Recall Ceiling:

A Null Result for Hierarchical Retrieval on Multi-Hop Document QA

Yu-Ting Lee
Apertis AI
leo.lee@apertis.ai

Code & data: <https://github.com/apertis-ai/multihop-doc-qa>

May 8, 2026

Abstract

We test whether hierarchical text retrieval improves over flat dense retrieval on multi-hop scientific document QA. On $n = 500$ multi-hop DocHop-QA (Park et al., 2025) questions, HiRAG using the **same encoder at the document and section stages** ties flat dense retrieval on every metric we measured (paired Wilcoxon $p = 0.79$; Cohen’s $d = -0.016$, a **negligible effect**; 345 / 500 questions receive bit-identical judge scores).

We diagnose the cause as a **0.15 absolute gap between document-level and section-level recall@5** that is invariant across three independent retriever families (jina dense, OpenAI dense, BM25 sparse) and trace it to a lexical-overlap bias: gold sections that dense retrieval *hits* at top-5 have +35 % higher question-section Jaccard than gold sections it *misses* (Welch $t = 13.15$ on $n = 1,985$ query-section comparisons, $p \ll 0.001$).

A second benchmark converges. On M3DocVQA (Cho et al., 2024) ($n = 464$ multi-page PDFs, an entirely different modality and domain), we replace a heuristic typed-edge graph-traversal policy (D1) with an LLM-guided per-hop traversal policy (LG1, *claude-sonnet-4-6* routing each hop) and find LG1 is **statistically significantly worse under both answer LLMs we tested**: $\Delta = -5.17$ pt under *qwen3.6-plus* (paired McNemar (McNemar, 1947) $p = 0.015$) and $\Delta = -9.05$ pt under *claude-sonnet-4-6* ($\chi^2_{cc} = 14.746$, $p = 0.000104$), same direction, two orders of magnitude more significant under the stronger LLM. A cross-LLM decomposition rules out the “qwen is the weak link” reading: a stronger answer LLM rescues D1 (+5.39 pt, $p = 0.006$) but cannot rescue LG1 (+1.51 pt, $p = 0.54$, n.s.). We frame the result as a **statistically significant cross-dataset null result for retrieval-side re-architecting**, held under two answer LLMs, in the methodological spirit of recent work on null results and statistical power in NLP (Card et al., 2020).

1 Introduction

A standard intuition in retrieval-augmented question answering: hierarchical structure helps. If the answer requires reasoning over multiple passages of multiple documents, the natural pipeline is to shortlist documents first (the easier discriminative problem) and then re-rank passages within those documents (the harder localisation problem). Several recent systems codify this intuition: HiRAG, GraphRAG, LightRAG, and their descendants. They report consistent gains over flat dense baselines on multi-hop benchmarks.

We re-test the intuition under three controls that prior multi-hop document QA work has not applied jointly:

1. **A fixed encoder at both stages** of the hierarchy, so any measured improvement is attributable to the *hierarchical structure* itself, not to a more expressive section encoder smuggled in at the second stage.

2. **A section-level retrieval metric** in addition to the standard document-level metric, so that the system’s actual passage-retrieval performance is observable rather than inferred from the end-task accuracy.
3. **A cross-vendor judge sanity check**, in which a 30 % subsample of system outputs is re-scored by a second judge from a different vendor to bound single-judge bias.

All comparisons follow standard reporting practice — paired statistics on per-query outcomes, multiple-comparison correction, paired effect sizes (Cohen’s d), and a priori power calculations following (Card et al., 2020); full statistical methodology in §3.6.

Under these controls, on 500 multi-hop questions from DocHop-QA, hierarchical text retrieval (HiRAG) provides **no measurable lift** over flat dense retrieval. Both systems hit the same retrieval set within metric precision (doc $r@5 = 0.664$; sec $r@5 = 0.514$), and 345 of 500 questions receive *exactly the same* judge score from both systems. Our diagnosis is concrete: the bottleneck is not hierarchical reasoning but section retrieval. Both systems hit the right paper 66 % of the time and the right section only 51 % of the time. The 15-percentage-point gap between document and section recall is **invariant across three independent retriever families** — jina-embeddings-v5-text-small (dense, 1024 d), text-embedding-3-small (dense, 1536 d), and BM25Okapi (sparse, no learned representation). **The section-recall ceiling we measure defines the empirical target any future text-only intervention on this benchmark must clear, and the lexical-overlap mechanism we identify points to *what* such an intervention must change.** We do not survey closure techniques; that is the contribution of follow-up work and is empirically incremental rather than architectural.

1.1 Contributions

We make **one primary contribution** (a measured section-recall ceiling), supported by **four robustness findings** and one reframing.

Primary: section-recall ceiling on multi-hop DocHop-QA. We measure section-level recall@5 against a (pmc_id, section, subsection) composite key — the finest granularity at which DocHop-QA publishes gold passage labels — and find that this metric runs **0.15–0.17 below document-level recall@5 across three retriever families** (§4.2, §4.6). The gap is the binding constraint on text-only multi-hop document QA on this benchmark.

Robustness 1 — HiRAG ties flat dense retrieval. On $n = 500$, HiRAG and flat dense retrieval are operationally identical (paired Wilcoxon $p = 0.79$; Cohen’s $d = -0.016$, negligible; 345/500 bit-identical judge scores). The tie holds under BM25 ($p = 0.95$, §4.6) and under a *claude-opus-4-6* cross-judge sanity check (§4.7).

Robustness 2 — Section-recall ceiling invariant across retriever families. Section $r@5$ holds at 0.514 across two dense families (jina-v5-small, text-embedding-3-small) and the BM25 doc-vs-section gap (0.166) tracks the dense gap (0.150) within 1.6 pp (§4.6). Two dense + one sparse family are sufficient to identify the ceiling as a property of the benchmark, not of any encoder.

Robustness 3 — Mechanism: lexical-overlap bias. Gold sections dense retrieval *hits* at top-5 have mean Jaccard 0.0869 versus 0.0566 for *missed* gold sections (+0.030 gap; Welch $t = 13.15$ on $n = 1,985$ query-section comparisons, $p \ll 0.001$, §5.1). Dense systematically misses low-overlap-but-evidential sections (buried results paragraphs, table cells, methods sub-sections).

Robustness 4 — Cross-dataset convergence on M3DocVQA. On $n = 464$ multi-page PDF questions, replacing heuristic typed-edge graph traversal (D1) with LLM-guided per-hop traversal (LG1) is **statistically significantly worse under both answer LLMs we tested**: $\Delta = -5.17$ pt under *qwen3.6-plus* ($p = 0.015$) and $\Delta = -9.05$ pt under *claude-sonnet-4-6* ($p = 0.000104$, §4.8). A cross-LLM decomposition rules out a “weak qwen” confound (a stronger reader rescues D1 by +5.39 pt but cannot rescue LG1, +1.51 pt n.s., §4.8.1). The 7-pt per-query oracle headroom that remains is not aligned with any structural query feature we tested.

Reframing — IRCOT. Iterative retrieval is cost-dominated on a strong dense base (mean 3.38 vs. dense 3.48), but on a weak (BM25) base it becomes the **best runner** (mean 3.49 vs. dense 3.03). The “strictly cost-dominated” reading of v2 was retriever-dependent.

Reproducibility artefact. A lightweight, fully-cached experimental pipeline runs end-to-end on a laptop (CPU only) given a single API key, in ~110 min wall clock at total \$195 LLM cost (including the cross-judge sanity check). Code, judge cache, and configs are released at <https://github.com/apertis-ai/multihop-doc-qa>.

1.1.1 1.2 What this paper does *not* claim

We do not claim that all hierarchical retrieval is useless. We claim it is useless **when the section-ranking stage uses the same encoder as the document-ranking stage**. A second-stage reranker that uses a *different* model — a cross-encoder, a late-interaction retriever, or a generation-time LLM rerank — is a different system that we do not test here.

We do not claim that the 0.15–0.17 doc-vs-section recall gap cannot be closed. The gap *is* an improvable quantity. This paper identifies and mechanistically explains it; closing it is the next paper.

We test layout-aware retrieval only on a cross-dataset setting, and only via a specific comparison: heuristic typed-edge BFS vs LLM-guided per-hop traversal. The cross-dataset experiment in §4.8 is run on M3DocVQA rather than DocHop-QA (the DocHop-QA corpus has no layout JSON available; M3DocVQA ships visually rich PDFs that docling can parse into typed-edge layout graphs). We do *not* claim that this single comparison exhausts the design space of layout-aware retrieval systems. In particular, we do not test: (a) late-interaction section retrievers (ColPali / ColBERT) *as retrievers* on DocHop-QA — we use ColPali only as the shared shortlist inside the M3DocVQA ablation; (b) passage-level rerankers; (c) learned traversal policies trained on a routing objective. The §4.8 result is that the specific form “LLM-guided per-hop traversal over a fixed graph” does not yield a uniform improvement over its heuristic counterpart, and is statistically significantly worse on the tested mix. A learned router or a different traversal target could still close the ceiling; that is future work.

We do not claim that BM25 is competitive with dense retrievers in absolute terms — it is not (mean 3.03 vs 3.48, a 0.45 deficit). We claim that BM25’s section-recall *gap* (doc r@5 – sec r@5) reproduces the dense gap to within 1.6 pp, which is the relevant quantity for the ceiling claim. The lexical-overlap mechanism in §5.1 partially explains this: BM25 is *more* lexically biased than dense by construction yet hits the same section-level ceiling, indicating that lexical bias is one mechanism of several, not the sole driver.

1.2 Roadmap

§2 reviews related work on multi-hop document QA, hierarchical retrieval, and the IR literature on document-vs-passage retrieval gaps. §3 describes the dataset, baselines, scoring protocol, statistical methodology, and the section-id construction. §4 reports the headline null result, the section-recall ceiling, the per-type slice analysis, the cross-encoder robustness check, the cost analysis, **the BM25 sparse baseline (§4.6), the cross-vendor judge sanity check (§4.7), the cross-dataset layout-aware-traversal ablation on M3DocVQA (§4.8), and the cross-LLM replication of that ablation with claude-sonnet-4-6 (§4.8.2).** §5 discusses the mechanism (now empirically supported), the ceiling as a measurable target for follow-up work, prior IR work that motivates measuring section recall, and limitations. §6 concludes.

2 Background and related work

2.1 Multi-hop document QA benchmarks

Multi-hop QA has been studied extensively at the open-domain level (HotpotQA (Yang et al., 2018), 2WikiMultiHopQA (Ho et al., 2020), MuSiQue (Trivedi et al., 2022)) and more recently at the document level, where the retrieval unit is a long scientific paper or technical report rather than a Wikipedia paragraph (M3DocVQA (Cho et al., 2024), MultiHop-RAG (Tang and Yang, 2024), DocHop-QA (Park et al., 2025)). Document-level multi-hop is harder than open-domain multi-hop because the unit of evidence is a passage *inside* a document — a section, a table cell, a figure caption — and the retrieval system must localise it, not just identify the right document.

DocHop-QA (Park et al., 2025) is the benchmark we use. It is a multi-hop scientific QA dataset (~11 K records) covering both single-document (1-hop) and multi-document (2-hop) questions, with two question concept types: “Paragraph-Oriented” (text-bearing) and “Table-Oriented” (table-bearing). The published release ships questions plus gold contexts (PMC-id-tagged passages) but **no gold answer field**. We discuss our scoring response in §3.4.

2.2 Hierarchical and graph-augmented retrieval

A line of recent retrieval pipelines builds explicit hierarchy or graph structure on top of dense retrieval. **HiRAG** (Huang et al., 2025) uses a two-stage doc-then-section pipeline with the same encoder at each stage. **GraphRAG** (Edge et al., 2024) and **LightRAG** (Guo et al., 2024) build entity and relation graphs over the corpus and traverse them at query time — a different mechanism from HiRAG’s hierarchy. **ColBERT** (Khattab and Zaharia, 2020) and its descendants — including **ColPali** (Faysse et al., 2024) for visual document retrieval — use late-interaction at the token level to recover fine-grained passage relevance that bag-of-vectors dense retrievers lose, which is again a different mechanism from same-encoder hierarchy. **monoT5** (Nogueira et al., 2020) and **RankT5** use a separately trained reranker on top of a dense retriever, also distinct from hierarchical retrieval.

These four lines of work are sometimes lumped under the slogan “structure helps multi-hop retrieval”, but they invoke *different* notions of structure: encoder hierarchy (HiRAG), graph traversal (GraphRAG, LightRAG), token-level interaction (ColBERT), and cross-encoder rerank (monoT5). This paper isolates the *first* of these four — same-encoder hierarchy — and reports a null result for that specific architectural variant. The other three remain unaddressed by our experiments and we do not draw conclusions about them.

2.3 The retrieval-vs-reasoning ablation gap

Most prior multi-hop document QA papers report two retrieval numbers: end-task accuracy and document-level recall. Section-level or passage-level retrieval is rarely measured against ground truth, partly because benchmarks rarely publish passage-level annotations.

The IR community has long understood that bag-of-vectors dense retrievers miss fine-grained passage relevance — this is the explicit motivation for late-interaction retrieval (Khattab and Zaharia, 2020), neural passage rerankers (Nogueira et al., 2020), and hierarchical embedding objectives such as Matryoshka representation learning (Kusupati et al., 2022). What the IR literature has not done — and what this paper contributes — is **measure the document-vs-section recall gap on a multi-hop document QA benchmark, against gold passage annotations the benchmark itself publishes, with paired statistics across runners and across three independent retriever families**.

We position this paper’s section-level recall metric as a **measurement** of a phenomenon long understood qualitatively in IR, not as the discovery of the phenomenon. The novelty is in the quantitative measurement on DocHop-QA, the demonstration of constant gap size across runners, retriever families, and judge vendors, the empirical mechanistic test in §5.1, and the

use of the gap to explain a null result on hierarchical retrieval that prior multi-hop QA work would otherwise have to attribute to “the reasoner”.

3 Methods

3.1 Dataset and slicing

DocHop-QA contains 11,379 records. We restrict to the multi-hop slice (`used_doc == "multi"`, $n = 7,863$) for all experiments because the single-document subset is trivially solvable by extractive reading and does not exercise the retrieval component.

Three pilot slices, drawn with a fixed seed (42), provide the experimental data:

- **$n = 500$ mixed multi-hop**, the headline pilot: 328 text-bearing + 172 table-bearing questions. Used for the dense-vs-hirag null result, both retrieval metrics, and rerun in full under BM25 for §4.6.
- **$n = 100$ multi-hop table-only**, with two paired runs using the jina and OpenAI encoders. Used for the encoder-swap robustness check in §4.4.
- A 100-question mixed-multi-hop pilot with the jina encoder, used as a cross-pilot consistency check.
- **$n = 50$ random subset of the $n = 500$ mixed pilot** $\times 3$ runners ($n = 149$ valid re-scores after parse failures), re-judged with `claude-opus-4-6` for the cross-vendor sanity check in §4.7.

All slices are seed-locked. The `(seed, n, query_type)` tuple deterministically reproduces the same question set across runs.

3.2 Baselines

We test three text-only baselines, all of which share the same retriever, the same answer LLM (`gpt-5.4-mini` via the Apertis gateway), and the same downstream judge. The differences are purely in the retrieve-then-answer logic.

DenseAnswerRunner is the reference baseline: a single embeddings call against the document corpus, top- $k = 6$ sections retrieved, and a single LLM answer generated against those sections.

HiRAGRunner (Huang et al., 2025) is the hierarchical text retriever. Stage 1: embed every section in the corpus, score each document by the mean similarity of its sections, and shortlist the top doc-top- $k = 5$ documents. Stage 2: re-rank the sections inside those shortlisted documents and take the top section-top- $k = 6$. Stage 3: generate the answer over those top sections. The encoder is the same at both stages.

TextIRCoTRunner is an iterative retrieve-and-reason baseline based on IRCoT (Trivedi et al., 2023): up to `max_steps = 3` iterations, each one issues a fresh sub-question and a fresh retrieval, with deduplication across iterations. v2 of this paper called IRCoT “strictly cost-dominated”; §4.6 shows that this reading was retriever-dependent — under BM25, IRCoT becomes the *best* runner of the three. We discuss the reframing in §4.6.

3.3 Retrievers

We test **three retriever families** in this paper:

- **jina-embeddings-v5-text-small** (dense, 1024 d), the default for $n = 100$ paired ablations. L2-normalised cosine similarity.
- **text-embedding-3-small** (dense, 1536 d, OpenAI), used as the headline retriever for the $n = 500$ mixed pilot. L2-normalised cosine similarity.

- **BM25Okapi** (Robertson and Zaragoza, 2009) (sparse, no learned representation, `rank_bm25` library defaults), used in the $n = 500$ BM25 rerun (§4.6). Tokenisation: lowercased alphanumeric runs (`[a-z0-9]+`), no stopword removal, no stemming.

Both dense encoders are invoked in batches of up to 64 inputs and are cached on disk by SHA-256 fingerprint of corpus content $\times$ encoder model id. BM25 indexing runs in ~ 10 s on a laptop and is not cached. Re-running an experiment is free on cache hit for the dense families.

3.4 Scoring protocol

DocHop-QA’s public release does not ship gold answers. We adopt a two-track scoring strategy.

LLM-as-judge (primary, generation quality). For each question + system answer + gold-context bundle, we prompt `gpt-5.4-mini` with a strict 0–5 rubric. The judge is given the question, the system’s answer, and the trimmed `gold_contexts` as evidence. Each judge call is cached on disk by a hash of (model, question, answer, contexts, prompt prefix). We report mean judge score and $\text{pct} \geq 4$ as headline numbers.

The LLM-as-judge protocol overlaps in spirit with G-Eval (Liu et al., 2023) and GPTScore (Fu et al., 2023); we adopt their model-graded approach with the caveat that we cache and pin the judge model version for reproducibility, and that we run the cross-vendor sanity check in §4.7 to bound single-judge bias.

We additionally run a **relaxed-rubric sensitivity pass** on every query that scored ≤ 2 under the strict rubric.

Retrieval metrics (secondary, deterministic, in two granularities). At each query we compute recall@k , precision@k , and nDCG@k for $k \in \{1, 5, 10\}$, plus MRR. We compute these at two granularities:

- **Document level:** predicted ids and gold ids are both `pmc_id` strings.
- **Section level:** predicted ids and gold ids are both composite keys of the form `f"{pmc_id}||{section}/{subsection}"`. Both sides of the comparison use this convention.

The section-level measurement is, to our knowledge, new for DocHop-QA. It is the central diagnostic of the paper.

3.5 Section-id construction

A section is a (`pmc_id`, `section`, `subsection`) triple. We flatten it into a composite key in two steps: join `section` and `subsection` with `/`, then join that to `pmc_id` with `||`. Both sides of the comparison use this convention.

3.6 Statistical methodology

For each runner we compute:

- **Bootstrap percentile 95 % CIs** for the mean judge score ($B = 2,000$ resamples, seed 42).
- **Paired Wilcoxon signed-rank tests** for runner-vs-runner comparisons, paired on `query_id`.
- **Paired effect sizes** (Cohen’s d , computed as $\text{mean_diff} / \text{sd_diff}$ over all 500 questions, treating tied pairs as zero-difference observations).
- **Paired McNemar tests** (McNemar, 1947) for binary-accuracy comparisons (used throughout §4.8 and §4.8.1 on M3DocVQA paired system outputs), with continuity correction and the exact two-sided binomial on the discordant pairs as a sanity check.

- **A priori power:** at $n = 155$ non-tied paired observations and $\alpha = 0.05$, the paired Wilcoxon test (using its normal-approximation efficiency ≈ 0.955 relative to the paired t) is powered to detect a Cohen’s d of approximately **0.225** at 80 % power. Our methodological stance on reporting null / negative results follows (Card et al., 2020): a well-powered non-significant comparison is itself informative, provided the power calculation is explicit.

On multiple comparisons. §4 reports approximately 14 paired-Wilcoxon p-values across runner-vs-runner, per-slice, BM25, and cross-judge tests. None of the headline claims rest on a single p-value crossing 0.05: the dense-vs-hirag tie ($p = 0.79$) is robust under any correction; the IRCOT regression on the text slice ($p = 0.04$ uncorrected) becomes marginal under Bonferroni for 14 tests ($p_{\text{corrected}} \approx 0.56$) and we flag it accordingly in §4.3 and §4.6.

We deliberately *separate* “no significant difference” from “tied within metric precision” in the results discussion. The dense-vs-hirag headline is the latter: 345 / 500 questions produce exactly the same judge score from both systems.

4 Results

4.1 The headline null: dense $\equiv$ HiRAG on $n = 500$ multi-hop

Table 1: Headline judge and retrieval metrics, $n = 500$ mixed multi-hop, text-embedding-3-small embedder. Bootstrap 95 % CIs in brackets.

runner	mean	95 % CI	pct ≥ 4	doc r@5	doc nDCG@10	sec r@5	sec nDCG@10
dense	3.484	[3.400, 3.574]	48.4 %	0.664	0.682	0.514	0.573
hirag	3.494	[3.408, 3.582]	49.0 %	0.664	0.681	0.514	0.573
text_ircot	3.382	[3.296, 3.472]	47.2 %	0.640	0.656	0.476	0.533

The dense and hirag rows are tied within metric precision on every retrieval metric. Document-level recall@5 is 0.664 for both; section-level recall@5 is 0.514 for both; document and section nDCG agree to one part in a thousand. This is not a small difference that fails to reach significance — it is *no difference at all* at the precision the metric can express.

The judge score difference is **0.010** in HiRAG’s favour, $p = 0.79$, Cohen’s d = **−0.016**. Of the 500 questions, 345 receive exactly the same judge score from dense and hirag. On the 155 non-tied pairs the mean difference is -0.032 (slightly favouring hirag) with paired Wilcoxon $z = -0.27$, $p = 0.79$ and a Cohen’s d of -0.016 across all 500 paired observations — an order of magnitude below the $n = 155$ minimum detectable effect of $d \approx 0.225$ at $\alpha = 0.05$, 80 % power.

Detailed paired Wilcoxon with effect sizes.

comparison	n_diff	mean_diff	Cohen’s d	z	two-sided p (uncorr.)	Bonferroni- adj p (k=14)
dense vs hirag	155	−0.032	−0.016	−0.27	0.789	1.000
dense vs text_ircot	253	+0.202	+0.106	−1.75	0.080	1.000

comparison	n_diff	mean_diff	Cohen’s d	z	two-sided p	Bonferroni-
					(uncorr.)	adj p
hirag vs text_ircot	234	+0.239	+0.117	−1.94	0.052	(k=14) 0.728

4.2 The section-recall ceiling

runner	doc r@5	sec r@5	gap
dense	0.664	0.514	0.150
hirag	0.664	0.514	0.150
text_ircot	0.640	0.476	0.164

The gap is 0.150 absolute, constant across runners. When a DocHop-QA multi-hop question has its evidence in the right paper, the right *passage* is recovered just over half the time. There is a 15-percentage-point absolute headroom that lives entirely inside correctly-retrieved papers and that no choice of runner among the three we tested can move. §4.4 shows that no choice of dense encoder moves it either, and §4.6 extends the result to a sparse retriever family.

This is the actual lever for multi-hop document QA on this benchmark. Any future system that wants to improve mean judge score has to lift section recall, not document recall.

Bimodal failure structure. A per-question breakdown of dense’s section recall reveals a bimodal failure pattern rather than a smooth distribution. Of the 248 / 500 questions with perfect document recall (doc r@5 = 1.0), **33 (13 %)** drop to sec r@5 < 0.5 — i.e., the right paper is retrieved but the right section is not. The section-recall histogram has clear clusters at 0.0 (92 queries) and 0.8–1.0 (172 queries), with relatively few intermediate cases. This suggests that “section ceiling failure” is a categorical property of the question’s evidence structure (where in the paper the multi-hop fact lives), not a smooth function of question difficulty. We return to this in §5.1 with the lexical-overlap mechanism.

4.3 Per-type slice: text vs table

Splitting the 500 mixed pilot by `query_type`:

Text questions (n = 328, 65.6 %)

runner	mean	95 % CI	pct ≥ 4
dense	3.692	[3.598, 3.790]	61.3 %
hirag	3.686	[3.588, 3.780]	60.7 %
text_ircot	3.555	[3.457, 3.655]	57.6 %

Table questions (n = 172, 34.4 %)

runner	mean	95 % CI	pct ≥ 4
dense	3.087	[2.930, 3.244]	23.8 %
hirag	3.128	[2.983, 3.285]	26.7 %
text_ircot	3.052	[2.884, 3.215]	27.3 %

The table slice is uniformly the harder slice. Every runner drops 0.55–0.60 mean from text $\rightarrow$ table and pct ≥ 4 collapses from $\sim 60\%$ to $\sim 25\%$.

The IRCOT regression on text is significant uncorrected ($p = 0.04$) but marginal after Bonferroni correction. We report the directional finding and flag it as not surviving correction. (We revisit IRCOT in §4.6 — the “cost-dominated” reading was retriever-dependent.)

4.4 The encoder swap: jina vs OpenAI dense

We test whether the section-recall ceiling can be moved by switching the underlying dense encoder. We use a **paired** $n = 100$ multi-hop table-only slice.

runner	mean (jina)	mean (OpenAI)	mean_diff	z	two-sided p
dense	3.16	3.31	+0.294	−1.19	0.234
hirag	3.21	3.33	+0.235	−1.05	0.292
text_ircot	3.23	3.17	−0.115	−0.44	0.659

Section recall does not move. On the same 100 table questions, section-level recall@5 is 0.54 with jina and 0.53 with OpenAI. The 0.150 ceiling is **robust** across the two dense encoders.

None of the LLM-judge differences across encoders reach statistical significance at $n = 100$; we *do not* claim that OpenAI is better or worse than jina as a general retriever.

4.5 Cost analysis

runner	answer \$ ($n = 500$)	judge \$ ($n = 500$)	combined \$/query
dense	\$17.41	\$19.10	\$0.073
hirag	\$18.38	\$19.03	\$0.075
text_ircot	\$29.79	\$18.04	\$0.096

Under dense retrieval, ircot’s answer cost is 70 % higher than dense’s because each iteration step issues a fresh LLM call, with quality tied or worse. **§4.6 shows this cost-dominance reverses under BM25.** The total cost of all pilots in this paper is **\$195 LLM** ($n = 500$ mixed dense + $n = 500$ mixed BM25 + $n = 100$ table jina + $n = 100$ table OpenAI + $n = 149$ cross-judge re-score), with no GPU component.

4.6 The BM25 sparse baseline: a third retriever family

To strengthen the encoder-robustness story from “two dense encoders” to “three independent retriever families”, we re-run the full $n = 500$ mixed pilot with **BM25Okapi** as the retriever, holding everything else identical (same questions, same answer LLM, same judge cache key with retriever as part of the fingerprint).

Table 2: $n = 500$ mixed multi-hop under BM25 retrieval. Same questions, same baselines, same judge as Table 1.

runner	mean	pct ≥ 4	doc r@5	sec r@5	sec nDCG@10
dense (BM25)	3.028	48.8 %	0.552	0.386	0.431
hirag (BM25)	3.046	48.2 %	0.552	0.386	0.432

runner	mean	pct ≥ 4	doc r@5	sec r@5	sec nDCG@10
text_ircot (BM25)	3.486	60.2 %	0.571	0.385	0.431

Paired Wilcoxon under BM25:

comparison	n_diff	mean_diff	Cohen’s d	z	two-sided p
dense vs hirag	270	−0.033	−0.012	−0.07	0.946
dense vs text_ircot	289	−0.792	−0.312	−6.44	< 0.001
hirag vs text_ircot	310	−0.710	−0.299	−6.33	< 0.001

Three findings.

(1) The dense-vs-hirag tie generalises to a sparse retriever. Under BM25, dense and hirag are again statistically and operationally identical ($p = 0.95$, Cohen’s $d = -0.012$). The HiRAG hierarchy adds nothing on top of a BM25 base either. Combined with §4.1 and §4.4, the null result holds across **three independent retriever families**.

(2) The doc-vs-section-recall gap is invariant across retriever families.

retriever family	doc r@5	sec r@5	gap
jina dense (n = 100 table)	0.61	0.54	0.07†
OpenAI dense (n = 500 mixed)	0.664	0.514	0.150
BM25 sparse (n = 500 mixed)	0.552	0.386	0.166

†The jina row is not directly comparable in absolute terms because it is the table-only slice; it is shown for completeness. The two $n = 500$ mixed rows (OpenAI dense and BM25 sparse) are directly paired and their gap differs by **only 1.6 pp** on a substantively different retriever family. The section-recall ceiling is a property of the benchmark’s evidence-distribution structure, not of any single encoder family.

(3) IRCOT reverses sign under BM25 — and is the best runner. Under dense retrieval (Table 1), IRCOT’s mean was 3.38, slightly *below* dense ($p = 0.08$). Under BM25 (Table 2), IRCOT’s mean is **3.49** — *above* both dense (3.03) and HiRAG (3.05), with paired $d = -0.31$, $p < 0.001$ in IRCOT’s favour. The “strictly cost-dominated” reading of v2 was retriever-dependent: when the base retriever is strong, additional iterations are redundant; when the base retriever is weak, iterative retrieval recovers a substantial part of the gap (mean +0.46 over dense). This is the **value-on-weak-retriever pattern** for iterative retrieval, and it has been understated in v2 of this paper.

The cost picture also reverses: at \$0.096 / query, IRCOT under BM25 produces +0.46 mean improvement, a **cost-effective gain** under BM25 (vs the same \$0.096 producing −0.10 mean change under dense). The implication for practitioners: do not pay for IRCOT on top of a strong dense retriever, but iterative retrieval is the cheapest available improvement on top of a sparse retriever.

The headline contribution of the paper — the section-recall ceiling and the dense $\equiv$ hirag null result — is *not affected* by this reframing. If anything, it is strengthened: the null holds under both retriever paradigms.

4.7 Cross-vendor judge sanity check

The v2 of this paper listed “single judge model” as a top-three limitation. We close it here by re-scoring **n = 50 random questions** $\times$ **3 runners** = **150 candidate scores** (149 valid

after one parse failure) with `claude-opus-4-6` (Anthropic) using the same strict rubric and gold contexts.

Table 3: cross-vendor judge agreement on a 50-question random subset.

runner	n	mean (mini)	mean (claude)	mean_diff	Pearson r	Spearman ρ	mean abs diff
dense	49	3.37	3.59	+0.22	0.546	0.429	0.63
hirag	50	3.36	3.66	+0.30	0.328	0.291	0.86
text_ircot	50	3.32	3.84	+0.52	0.680	0.549	0.72
overall	149	3.35	3.70	+0.35	0.505	0.411	0.74

Two observations.

(1) **Claude is systematically more generous, with a consistent +0.35 offset, but rank-ordering is preserved.** Pearson 0.51 across $n = 149$ paired observations is moderate-to-strong agreement at this sample size; Spearman 0.41 reflects judge-induced compression of the 0–5 score range. Critically, **Claude does *not* rank dense differently from HiRAG:** Claude’s dense mean is 3.59, Claude’s HiRAG mean is 3.66, a +0.07 differential — well below the $n = 50$ detection threshold for either rubric and roughly *the same direction and magnitude* as the mini’s -0.01 differential on the same subset. **The dense $\equiv$ HiRAG tie is not an artefact of the mini judge.**

(2) **Claude favours IRCOT more than the mini does.** Under the mini judge, dense beats IRCOT by 0.05 on this subset; under Claude, IRCOT beats dense by 0.25. Combined with the BM25 IRCOT result in §4.6, two independent signals (a different vendor judge + a different retriever family) suggest the v2 conclusion that “IRCOT is strictly cost-dominated” was over-stated and was an artefact of pairing IRCOT with a strong dense retriever and a single judge that under-weights its more elaborate evidence-gathering style.

Cost. The cross-judge sanity check cost \$25.45 (`claude-opus-4-6` is roughly $25\times$ the per-call price of `gpt-5.4-mini`). It is included in the \$195 paper total.

We continue to recommend pinned-judge protocols for reproducibility, but we no longer treat single-judge as an open limitation: the dense $\equiv$ HiRAG headline is robust across vendors at the magnitudes we observe.

4.8 Cross-dataset evidence: LLM-guided graph traversal on M3DocVQA

§5.2 lists layout-aware retrieval over typed document graphs as one plausible class of system that could close the section-recall gap. We test one specific form on a different multi-hop document QA benchmark and report a robustly negative result.

Setup. On 464 questions from M3DocVQA (visually rich multi-page PDFs, distinct from DocHop-QA in modality and domain), we hold every component fixed except the per-hop traversal policy. **D1** uses heuristic typed-edge BFS over a docling-parsed layout graph + spaCy KG overlay, seeded at the top ColPali-shortlisted node (doc recall@5 = 0.95). **LG1** uses identical retrieval and identical graph but delegates each hop’s next-node selection to *claude-sonnet-4-6*. Full graph construction, edge families, hop budget, and prompts are in App. G. We use M3DocVQA as an *independent* test of whether retrieval-side re-architecting helps, not as a second data point on the DocHop-QA ceiling.

Result, primary configuration (Sonnet 4.6 answer LLM). LG1 is statistically significantly *worse* than heuristic D1: $D1 = 0.3470$ vs $LG1 = 0.2565$, $\Delta = -9.05$ pt (paired McNemar $\chi^2_{cc} = 14.746$, $p = 0.000104$ on $n = 464$; 78 D1-only-correct vs 36 LG1-only-correct discordant pairs). The qwen3.6-plus replication on the same 464 questions reproduces sign and significance: $D1 = 0.2931$, $LG1 = 0.2414$, $\Delta = -5.17$ pt ($p = 0.015$). Full per-variant numerics, slice analyses, oracle bounds, and cross-LLM lift measurements are in App. G.

The answer-LLM choice is not the confound. The qwen→Sonnet 4.6 swap rescues D1 by **+5.39 pt** (paired $p = 0.006$) but rescues LG1 by only **+1.51 pt** ($p = 0.54$, n.s.). A stronger frontier reader extracts more signal from D1’s cleaner local neighbourhood, not from LG1’s LLM-wandered context. The paired-test gap therefore *widens* under the stronger LLM (5.17 → 9.05 pt) rather than shrinking, ruling out the “qwen is the weak link” reading.

Slice structure: no routing opportunity. Under Sonnet 4.6, no gold-doc-count slice has LG1 above D1 at statistical significance (1-doc $\Delta = -12.29$ pt, 2-doc $\Delta = -2.38$ pt). The qwen 2-doc slice +1.19 pt that v5 read as a routing signal *reverses* under Sonnet. A slice-gated oracle (gold-doc-count-aware) is +0.43 pt over D1 (NS, $p = 0.85$) under qwen and *negative* under Sonnet; a per-query oracle upper bound is **+7.11 pt under qwen and +7.76 pt under Sonnet** — robust across LLMs but not aligned with any *structural* query feature we tested. A learned router would have to recover the 7–8 pt headroom from content features. Cost: LG1 is $15\times$ D1 under qwen and $2.1\times$ under Sonnet; strictly cost-dominated under both.

What this adds to the headline story. Two retrieval-side interventions (DocHop-QA hierarchy in §4.1, M3DocVQA LLM-guided traversal here), on different modalities (text vs PDF), in the M3DocVQA case robust across two answer LLMs, both fail to beat their simpler baselines. The converging evidence strengthens the headline: **at the current multi-hop document QA benchmark difficulty, retrieval-side re-architecting is not where improvements come from.** The constructive candidate that survives is representation-level change (late-interaction retrievers, learned passage rerankers) — *what* is matched, not *how* the structural search proceeds. The conditional-routing hypothesis v5 derived from the qwen 2-doc slice does not survive the Sonnet replication and is withdrawn.

5 Discussion

5.1 Why is the section-recall gap there? An empirical mechanism

The mechanism. Dense embeddings reward sections whose surface text is most lexically similar to the question phrasing, regardless of whether those sections actually contain the multi-hop evidence. The abstract, the introduction, and the conclusion of a paper typically share the most lexical overlap with a question about that paper, so they tend to rank highest under a dense encoder. The actual evidence for a multi-hop fact is often elsewhere: in a table, in a buried methods sub-section, in a results paragraph that talks about specifics rather than restatements of the paper’s main claim. Those sections do not lexically echo the question’s framing, so they do not rise to the top of the dense ranking even when the document is correctly shortlisted.

The empirical test. v2 of this paper offered the lexical-overlap mechanism as a discussion-level hypothesis labelled “not tested in this paper”. We now report a direct measurement.

For each query in the $n = 500$ mixed pilot, we compute the question→section token Jaccard (lowercased alphanumeric tokens, English stopwords removed) for four groups:

group	n	mean Jaccard	median
gold sections (all)	1,985	0.0728	0.0610
gold sections hit by dense top-5	1,059	0.0869	0.0741
gold sections missed by dense top-5	926	0.0566	0.0490
dense top-5 retrieved (any)	2,500	0.0750	0.0625

Hit – Missed delta: +0.0303 mean Jaccard, **Welch $t = 13.15$** ($p \ll 0.001$) on $n = 1,985$ paired observations.

Three implications.

(1) Dense systematically misses low-overlap-but-correct sections. Gold sections that dense fails to retrieve at top-5 have ~35 % lower lexical Jaccard with the question than the

gold sections it does retrieve. This is a direct measurement of the lexical-overlap bias: dense’s relevance score is positively correlated with lexical overlap regardless of evidential content.

(2) Dense’s retrieval distribution closely tracks lexical overlap. Dense top-5’s mean Jaccard (0.0750) sits much closer to the *hit-gold* distribution (0.0869) than to the *missed-gold* distribution (0.0566). The retriever ranks by lexical overlap whether or not the section is evidential.

(3) The 0.150 doc-vs-section recall gap is at least partially explained by this mechanism. When a question’s gold section happens to lie in a low-overlap region of an otherwise correctly-retrieved paper (a buried results paragraph, a table, a methods sub-section), dense ranks it below higher-overlap-but-irrelevant sections from the same paper. The 33 / 248 perfect-doc-recall but low-section-recall queries identified in §4.2.1 are the population in which this mechanism is most clearly operating.

The BM25 caveat. A pure-lexical retriever (BM25 in §4.6) is by construction *more* lexical-overlap-biased than dense, yet hits a *higher* doc-vs-section gap (0.166 vs 0.150) and a substantially *lower* absolute section recall (0.386 vs 0.514). Lexical bias is therefore one mechanism in a stack — necessary but not sufficient — and the §5.1 hypothesis in its strongest form (“dense reduces to a smoothed lexical retriever”) is partially refuted. The weaker form (“dense’s section-level errors correlate strongly with low question-section lexical overlap”) is the form supported by Welch $t = 13.15$.

This is the strongest mechanistic statement we can responsibly make from these experiments. A full causal account would require either a learned retriever trained with a lexical-bias-removed objective, or attention-diagnostic ablations on a smaller open-source dense encoder. We leave both as future work.

5.2 The ceiling as a measurable target

The 0.15–0.17 doc-vs-section gap is what the lexical-overlap mechanism produces under bag-of-vectors retrieval. We do not survey closure techniques in this paper; closing the gap is the contribution of follow-up work, and is incremental and empirical rather than architectural. Companion work measures partial closure on DocHop-QA: a domain-adapted retriever closes 55.5 % of the gap, leaving 44.5 % unclosed at the encoder level. The ceiling we measure is therefore best read as a benchmark target: any future text-only intervention should be evaluated against it rather than against document-level recall, and any closure claim should disclose the absolute magnitude relative to this 0.15 reference. §4.8 tightens one corner of the design space — uniformly swapping heuristic graph traversal for LLM reasoning over a fixed typed-edge graph does not clear its local baseline on M3DocVQA, and no query-level gating we tested rescues the result (App. G.5) — but it does not foreclose other retrieval-side directions.

A natural reader hypothesis attributes the residual gap to *modality* — that tables and figures, encoded as flat text strings, lose structural signal that visual or layout-aware representations would recover. Two pieces of evidence in this paper push back against pure modality as the dominant mechanism. First, §4.3 shows that the dense $\equiv$ HiRAG tie holds on text-only queries ($n = 328$) at the same precision as on the mixed pilot, so the ceiling exists independent of table content. Second, §4.8 tests one form of multimodal-aware retrieval — ColPali shortlisting plus docling-parsed typed-edge layout graph plus LLM-guided traversal — and finds it significantly worse than the heuristic baseline under two answer LLMs. We therefore do not read the residual as a single-mechanism phenomenon; it likely combines modality, long-context representation, and compositional multi-hop demands, and we do not prescribe a closure technique.

5.3 Implications for prior work

The IR community has long known that bag-of-vectors dense retrievers underperform at passage-level relevance. Our contribution is not the discovery of this phenomenon — it is the **measure-**

ment of the gap on a multi-hop document QA benchmark, against gold passage annotations the benchmark itself publishes, with paired statistics across runners, **across three independent retriever families**, and **across two LLM judge vendors**, and a **direct empirical test of the lexical-overlap mechanism**.

The narrower implication for multi-hop document QA papers specifically: papers that report only document-level recall as their retrieval headline are missing 15+ percentage points of measurable retrieval performance. We recommend that future multi-hop QA work report both granularities, with section-level (or passage-level, when available) as the primary retrieval-quality metric.

5.4 Limitations

Single judge family. Headline numbers use `gpt-5.4-mini`; the §4.7 cross-vendor sanity check (`claude-opus-4-6`, $n = 149$ re-scored) preserves the dense $\equiv$ HiRAG tie (Claude +0.07, mini −0.01; Pearson 0.51) but is a 30 % subsample, not a full re-run.

Retriever family coverage. Three families tested (jina dense, OpenAI dense, BM25 sparse). We have not tested late-interaction retrievers (ColBERT-class) or passage rerankers (monoT5-class) *as retrievers* on DocHop-QA; either may exhibit a different ceiling.

Table-slice power. Of $n = 500$, only $n = 172$ are table queries; power is below 50 % at $d = 0.30$ and below 20 % at the observed $d = 0.04$. We do not claim the table-slice tie is significant; only the mixed-pilot tie.

Layout-aware retrieval scope. §4.8 tests one specific form (LLM-guided per-hop traversal vs heuristic typed-edge BFS over a docling + KG graph) on M3DocVQA, and finds it significantly worse under both tested answer LLMs (App. G). We do *not* test (a) late-interaction section retrievers as retrievers on DocHop-QA; (b) passage-level rerankers; (c) learned traversal policies trained on a routing objective; (d) a section-recall ceiling numerical replication on M3DocVQA (no public section-level gold labels in the split). The §4.8 result is bounded to the specific sub-question of LLM-guided traversal vs heuristic BFS over the same graph.

Cross-dataset answer-LLM and retrieval differ from the DocHop-QA headline. §4.8 uses ColPali + `claude-sonnet-4-6`; §4.1–§4.7 use jina/OpenAI dense or BM25 + `gpt-5.4-mini`. We describe §4.8 as a *cross-dataset convergence*, not a section-recall replication, and do not pool numerics. The answer-LLM concern is closed empirically by the qwen–Sonnet cross-variant measurements (App. G.4): the gap widens monotonically with answer-LLM strength, ruling out a weak-LLM artefact.

DocHop-QA’s withheld answer set. The benchmark ships gold contexts but not gold answers, forcing LLM-as-judge in place of EM/F1.

Mechanism is associative, not causal. §5.1 reports a +0.030 Jaccard gap (Welch $t = 13.15$, $n = 1,985$) between hit and missed gold sections — an associative measurement. A learned retriever trained to remove lexical bias would be the causal intervention. BM25’s section-recall ceiling reproducing the dense one (§5.1) further indicates lexical bias is one mechanism in a stack, not the sole driver.

6 Conclusion

We measure a section-recall ceiling on multi-hop scientific document QA. On $n = 500$ DocHop-QA questions, HiRAG with a same-encoder two-stage pipeline ties flat dense retrieval (paired Wilcoxon $p = 0.79$, Cohen’s $d = -0.016$, 345 / 500 bit-identical judge scores), and the doc-vs-section recall@5 gap holds at **0.150 across two dense retriever families and 0.166 under BM25** — invariant across three retriever families that span dense and sparse design choices.

We trace the gap to a lexical-overlap bias measurable at the section level: gold sections that dense retrieval *hits* at top-5 have +35 % higher question-section Jaccard than gold sections it

misses (Welch $t = 13.15$ on $n = 1,985$ query-section comparisons, $p \ll 0.001$). The mechanism is associative; a learned retriever that removes the bias would be the causal intervention.

A second benchmark converges. On M3DocVQA ($n = 464$ multi-page PDF questions), uniformly swapping heuristic typed-edge graph traversal for LLM-guided per-hop traversal is statistically significantly *worse* under both answer LLMs we tested ($\Delta = -9.05$ pt under *claude-sonnet-4-6*, $p = 0.000104$; $\Delta = -5.17$ pt under *qwen3.6-plus*, $p = 0.015$), with the gap *widening* under the stronger reader rather than shrinking — ruling out the “answer-LLM weakness” reading. The cross-benchmark lesson: across two datasets, two modalities, and two retrieval-side interventions (same-encoder hierarchy on DocHop-QA, LLM-guided graph traversal on M3DocVQA), retrieval-side re-architecting does not clear its local baseline.

A secondary finding: iterative retrieval (IRCoT) is not strictly cost-dominated. It is cost-dominated only when paired with a strong dense base; under BM25 it becomes the best runner of the three (mean 3.49 vs. 3.03 dense, $p < 0.001$), demonstrating that iterative retrieval performs meaningful compensatory work when the base retriever is weak.

The methodological recommendation we draw: **future work on multi-hop document QA should report section-level recall alongside document-level recall**. The former exposes ~ 15 percentage points of real retrieval headroom that the latter hides, the gap is invariant across three retriever families, and the lexical-overlap mechanism we identify is a measurable diagnostic for *why* a given retriever falls short. We make no architectural recommendations for closing the gap; that is the contribution of follow-up work.

7 Ethics statement

This paper reports a null result on a published scientific QA benchmark, using API-served dense encoders, a sparse open-source retriever, and two API-served LLMs (*gpt-5.4-mini* for the primary judge, *claude-opus-4-6* for the cross-judge sanity check). The benchmark is composed of public-domain biomedical literature (PubMed Central). We see no direct ethical risks from the empirical work itself.

We note one indirect risk: **judge-model lock-in**. We have partially mitigated this in v3 by running a cross-vendor sanity check; the mitigation is partial because the sanity check covers 30 % of the headline pilot, not 100 %. As LLM judges become standard in NLP evaluation, we recommend cross-vendor sanity checks as a routine reproducibility item.

We do not see the work as creating dual-use concerns or human-subjects risks beyond those already inherent in the source benchmark.

8 LLM use disclosure

This paper used Claude Opus 4.6 (1M context) for drafting assistance and for running multi-perspective writing reviews (sciwrite, ml-paper-writing, and nlp-engineer skill audits) and a Codex/GPT-5.4 simulated peer review. All experimental design, code implementation, data generation, statistical analysis, and final scientific judgements were authored by humans. The LLM did not access external data sources during drafting beyond the artefacts in our own research repository and (for citation verification) the public arXiv API.

8.1 Reproducibility statement

All code, configurations, run JSONs, judge cache, lexical-overlap analysis, and the per-query score breakdown are released alongside this paper. The pipeline runs on CPU (no PyTorch dependency): `scripts/run_text_experiment.py`, `src/baselines/text_runners.py`, and `src/retrieval/bm25_retriever.py` together let any reader re-run the full pilot suite on a laptop with an Apertis API key in approximately 110 minutes of wall clock and \$195 of LLM cost.

The judge cache is included so that re-running the analysis is free on cache hit. All commands and configs are in the README of the supplementary directory.

8.2 Acknowledgements

[redacted for review] ## Appendix A. Encoder, retriever, and answer-LLM inference parameters

- **jina-embeddings-v5-text-small**: dimension 1024, L2-normalised cosine similarity, batch size 64, accessed via Apertis gateway.
- **text-embedding-3-small**: dimension 1536, L2-normalised cosine similarity, batch size 64, accessed via Apertis gateway.
- **BM25Okapi** (`rank_bm25` library defaults): $k1 = 1.5$, $b = 0.75$, $\epsilon = 0.25$; tokeniser = lowercased `[a-z0-9]+` runs, no stopword removal, no stemming; index built once over the restricted-to-sample-docs corpus (~4k sections), build time ~10 s on laptop CPU.
- **gpt-5.4-mini (answer LLM)**: temperature 0.0, `max_tokens` 400 (dense / HiRAG) or 300 (IRCoT), system prompt in Appendix B.
- **gpt-5.4-mini (judge LLM)**: temperature 0.0, `max_tokens` 600 default, strict and relaxed rubric prompts in Appendix B.
- **claude-opus-4-6 (cross-judge LLM, §4.7)**: temperature 0.0, `max_tokens` 1024 (bumped from default to accommodate Claude’s more verbose JSON output), strict rubric only.

8.3 Appendix B. Judge prompts (strict and relaxed)

(Verbatim prompts from `src/eval/llm_judge.py` to be inserted here for the camera-ready version. The two prompts differ only in the introduction paragraph that establishes the rubric — strict treats the gold contexts as the only valid evidence; relaxed treats them as a starting set of evidence.)

8.4 Appendix C. Per-query score table

(Released as supplementary CSV. Includes `query_id`, `runner`, `retriever_family`, `judge_score`, `judge_vendor`, `strict_rationale`, `relaxed_score` (where applicable), `retrieved_doc_ids`, `retrieved_section_ids`, `runner_cost_usd`, `judge_cost_usd`.)

8.5 Appendix D. Power calculation

For a paired Wilcoxon signed-rank test at $\alpha = 0.05$ (two-sided), 80 % power, and asymptotic relative efficiency 0.955 vs the paired t-test, the minimum detectable Cohen’s d is approximately

$$d_{\min} \approx (z_{\{1-\alpha/2\}} + z_{\{1-\beta\}}) / \sqrt{n_{\text{diff}} \times 0.955}$$

At $n_{\text{diff}} = 155$ (the dense-vs-hirag headline comparison), $d_{\min} \approx 0.225$. The observed |Cohen’s d | of 0.016 is an order of magnitude below this threshold. At $n_{\text{diff}} = 270$ (the BM25 dense-vs-hirag comparison), $d_{\min} \approx 0.170$; observed $|d| = 0.012$. At $n_{\text{diff}} = 48$ (the table-slice dense-vs-hirag comparison), $d_{\min} \approx 0.41$ and the observed $|d| = 0.06$ is below threshold; the table-slice tie is power-limited and we do not claim it as significant.

8.6 Appendix E. Lexical-overlap analysis (full table)

Reproduced from `lexical_overlap_analysis.md`. Tokenisation: lowercased `[a-z0-9]+` runs, English stopwords removed.

group	n	mean Jaccard	median	std
gold sections (all)	1,985	0.0728	0.0610	0.0512
gold sections hit by dense top-5	1,059	0.0869	0.0741	0.0541
gold sections missed by dense top-5	926	0.0566	0.0490	0.0421
dense top-5 retrieved (any)	2,500	0.0750	0.0625	0.0498

Welch’s two-sample t (hit vs missed): $t = \mathbf{13.15}$, $df \approx 1947$, $p \ll 0.001$. Hedges’ $g \approx 0.61$ (medium-to-large effect).

8.7 Appendix F. Cross-vendor judge sanity check (full table)

Reproduced from `cross_judge_check.json`. Subset: 50 random questions from the $n = 500$ mixed pilot (seed = 42), 3 runners each, 1 parse failure dropped ($n = 149$ valid).

runner	n	mean (mini, strict)	mean (claude, strict)	Pearson	Spearman	Kendall τ	mean abs diff	claude cost
dense	49	3.37	3.59	0.546	0.429	0.382	0.63	\$8.49
hirag	50	3.36	3.66	0.328	0.291	0.252	0.86	\$8.30
text_ircot	50	3.32	3.84	0.680	0.549	0.486	0.72	\$8.66
overall	149	3.35	3.70	0.505	0.411	0.354	0.74	\$25.45

8.8 Appendix G. Cross-dataset layout-aware traversal experiment (§4.8)

This appendix records the full protocol, slice-by-slice numerics, mechanism analysis, and the cross-LLM defense for the §4.8 cross-dataset result.

8.8.1 G.1 Protocol (applies to both qwen and Sonnet 4.6 variants)

- **Retrieval shortlist:** ColPali top-5 over the full M3DocVQA corpus, doc recall@5 = 0.95.
- **Graph:** docling-parsed typed-edge layout graph per PDF, with spaCy-extracted entity-mention and entity-relation overlays. Four edge families: (i) layout reading-order, (ii) layout spatial-adjacency (same-page boxes within a geometric threshold), (iii) KG entity-mention (word $\rightarrow$ entity), (iv) KG entity-relation (entity $\leftrightarrow$ entity).
- **D1 traversal:** deterministic typed-edge BFS with fixed edge-priority ordering, seeded at the top-ColPali-scoring layout node, up to a fixed hop budget.
- **LG1 traversal:** per-hop next-node selection by `claude-sonnet-4-6` given the current evidence set plus candidate neighbours, confidence-based early stop, mean 3.3 LLM calls per query, up to 4 hops.
- **Answer generation:** fixed prompt template, temperature 0.0, max 256 output tokens. The two variants differ only in the answer LLM (`qwen3.6-plus` vs `claude-sonnet-4-6`); LG1’s per-hop traverser is `claude-sonnet-4-6` under both variants.

8.8.2 G.2 Per-variant numerics ($n = 464$)

Qwen-variant (replication). D1 acc = 0.2931; LG1 acc = 0.2414; $\Delta = -5.17$ pt. Paired McNemar: both correct = 79, neither = 295, D1-only = 57, LG1-only = 33; $\chi^2_{cc} = 5.878$, $p = 0.0153$; exact binomial on discordants $p = 0.0149$. Slice by gold-doc count: 1-doc ($n = 293$) 0.307 vs 0.225 ($\Delta = -8.19$ pt); 2-doc ($n = 168$) 0.262 vs 0.274 ($\Delta = +1.19$ pt, the signal eliminated under Sonnet); 3-doc ($n = 3$) 0.667 vs 0.000. Cost: D1 total \$0.22, LG1 total \$3.33 (15 $\times$ ratio).

Sonnet-variant (primary). D1-S acc = 0.3470; LG1-S acc = 0.2565; $\Delta = -9.05$ pt. Paired McNemar: both correct = 83, neither = 267, D1-S-only = 78, LG1-S-only = 36; $\chi^2_{cc} = 14.746$, $p = 0.000104$; exact binomial on discordants $p = 0.000104$. Slice by gold-doc count: 1-doc 0.386 vs 0.263 ($\Delta = -12.29$ pt); 2-doc 0.274 vs 0.250 ($\Delta = -2.38$ pt, reversed vs qwen); 3-doc 0.667 vs 0.000. By query type (Sonnet): TableQ 29.9 % vs 20.1 % ($\Delta = -9.76$ pt, $n = 164$); ImageListQ 32.4 % vs 13.5 % ($\Delta = -18.92$ pt, $n = 37$); TextQ 72.1 % vs 46.5 % ($\Delta = -25.58$ pt, $n = 43$); the only Sonnet slice where LG1 beats D1 is `Compose(ImageQ, TableQ)` at +5.36 pt ($n = 56$), small and non-significant. Cost: D1-S total \$1.69, LG1-S total \$3.60 ($2.1\times$ ratio).

Cross-LLM lifts on each arm. qwen→Sonnet lifts D1 from 0.2931 to 0.3470 (+5.39 pt, paired McNemar $\chi^2_{cc} = 7.481$, $p = 0.006$). qwen→Sonnet lifts LG1 from 0.2414 to 0.2565 (+1.51 pt, $p = 0.54$, n.s.). The asymmetry is the measurement underlying the body’s “stronger answer LLM rescues D1 but not LG1” finding.

8.8.3 G.3 Mechanism analysis

We hypothesise that on a 1-doc query the gold evidence is a few layout-adjacent nodes reachable by a 1–2-hop heuristic BFS: a table cell and its caption, a figure and its explanatory paragraph, or a text block and its adjacent heading. Heuristic BFS stays in this local region because its hop budget coincides with the local evidence neighbourhood. An LLM traverser, asked to keep hopping until it is “confident”, wanders 2–3 hops further into other pages of the *same document* — still on-domain, still topically similar, but not the gold evidence. The extra nodes dilute the evidence context, and the answer LLM hallucinates from the off-topic text. Under qwen, this same wandering occasionally appeared productive on 2-doc queries (+1.19 pt) and motivated a routing interpretation; under Sonnet 4.6 it is *not* productive (−2.38 pt), suggesting the qwen 2-doc signal was an artefact of qwen’s weaker synthesis ability rather than a genuine benefit of LLM-guided traversal. A stronger answer LLM compresses the amount of evidence needed to answer correctly, which makes D1’s cleaner local neighbourhood an even larger advantage.

8.8.4 G.4 Why the answer-LLM choice is not a confound

Four independent axes rule out the “qwen is the weak link” reading:

1. **Sign of the effect is identical and the stronger LLM widens the gap.** D1 beats LG1 by −5.17 pt under qwen and −9.05 pt under Sonnet, with a $\sim 125\times$ p-value drop. A weak-LLM confound predicts gap shrinkage under a stronger LLM; we observe the opposite.
2. **Stronger LLM rescues D1 but not LG1.** D1 lift +5.39 pt ($p = 0.006$); LG1 lift +1.51 pt ($p = 0.54$, n.s.). The answer-LLM-sensitive pathway is D1’s clean neighbourhood, not LG1’s wandered context.
3. **Mechanism is preserved.** The 1-doc regression is *larger* under Sonnet (−12.29 pt vs −8.19 pt); the 2-doc qwen +1.19 pt reverses to −2.38 pt; no Sonnet slice has LG1 above D1 at significance. The failure mode (LLM wandering into off-topic same-document pages) becomes more punitive as the answer LLM strengthens.
4. **Cost-performance asymmetry is robust.** LG1’s per-query cost is $15\times$ D1 under qwen and $2.1\times$ under Sonnet (the ratio compresses because Sonnet’s higher per-token price dominates both arms); the accuracy gap widens from 5.17 to 9.05 pt across the same swap. Under either LLM, LG1 is strictly cost-dominated.

A replication with a third answer LLM (e.g. gpt-5.4-mini matching the DocHop-QA headline) reversing the paired comparison would change our mind. Across the two LLMs we tested the comparison is robustly negative, monotonically worsens with stronger answer LLMs, and is structurally distinguishable from a simple weak-LLM confound. The paired McNemar is a consistent estimator of the retrieval-conditioned D1–LG1 delta for any fixed answer LLM because

the common downstream LLM call cancels on each paired query; the cross-LLM measurements above show that the actual cross-LLM behaviour aligns with the structural reading.

8.8.5 G.5 Oracle bounds on routing headroom

- (1) **Slice-gated oracle (gold-doc-count-aware).** Picks D1 on 1-doc and 3-doc, LG1 on 2-doc. Under qwen: $138/464 = 0.2974$, +0.43 pt over D1 (paired McNemar $\chi^2_{cc} = 0.036$, $p = 0.85$, n.s.). Under Sonnet: strictly *below* D1-S because the 2-doc Sonnet slice regresses to -2.38 pt. Coarse gold-doc-count gating fails under both answer LLMs.
- (2) **Per-query oracle upper bound.** Knows for every query whether LG1 or D1 will produce the correct answer. Under qwen: $169/464 = 0.3642$, +7.11 pt over D1-qwen. Under Sonnet: $197/464 = 0.4246$, +7.76 pt over D1-S (114 discordant pairs: 78 D1-S-only + 36 LG1-S-only). The headroom is slightly *larger* under the stronger answer LLM — robust across LLMs at $\approx +7-8$ pt over D1 standalone — but it is no longer aligned with any *structural* query feature. A learned router would have to recover the 7-8 pt from query content features (lexical, topical, retrieval-confidence cues).

8.8.6 G.6 Budget

M3DocVQA qwen variant: \$3.55 (D1 \$0.22 + LG1 \$3.33). Sonnet replication: \$5.29 (D1-S \$1.69 + LG1-S \$3.60). Total cross-dataset LLM spend: \$8.84. Paper-wide LLM total including DocHop-QA headline + cross-judge sanity check + cross-dataset variants: ~\$200, no GPU.

8.8.7 G.7 Released artefacts

Raw artefacts released with this paper (paths relative to the repo root):

- **Raw results (qwen variant, replication).** `rc-run/stage_21_layoutgraphrag_pilot/e_full_464.json` (E / LG1 full run, 464 paired records) and `d1_full.json` (D1 heuristic run, same 464 qids). Each record carries fields `qid`, `question`, `type`, `n_traversal_calls`, `traversal_cost`, `answer_cost`, `answer`, `gold`, `correct`.
- **Raw results (Sonnet 4.6 variant, primary headline).** `e_full_464_sonnet46.json` and `d1_full_sonnet46.json` — same 464 qids, identical retrieval + graphs + prompts, answer LLM swapped to `claude-sonnet-4-6` for both arms.
- **Ablation suite (qwen variant).** `b1_full.json`, `b2_text_only.json`, `b3_kg_only.json`, `c_layoutgraphrag.json`, `d1_full.json`, `d2_layout_only_edges.json`, `d3_kg_only_edges.json`, `d4_uniform_traversal.json` — eight Phase B/C/D configurations covering the H1/H2/H3 ablations. `summary.json` contains the corresponding pairwise contrasts.
- **Execution scripts.** `phase_e_llm_traversal.py` (qwen-variant LG1), `phase_d_ablations.py` (qwen-variant D1 and ablation suite), `phase_e_llm_traversal_sonnet46.py` (Sonnet-variant LG1), `phase_d_ablations_sonnet46.py` (Sonnet-variant D1). All four scripts use the same Apertis endpoint and the same ColPali top-20 shortlist (`retrieval_top20.json`).
- **Paired statistics, primary (Sonnet 4.6).** D1-S accuracy $161/464 = 34.70\%$, LG1-S accuracy $119/464 = 25.65\%$, $\Delta = -9.05$ pt. Paired McNemar on 114 discordants (36 LG1-S-only, 78 D1-S-only): $\chi^2_{cc} = 14.746$, **$p = 0.000104$** ; exact binomial on discordants $p = 0.000104$.
- **Paired statistics, replication (qwen).** D1 accuracy $136/464 = 29.31\%$, LG1 accuracy $112/464 = 24.14\%$, $\Delta = -5.17$ pt. Paired McNemar on 90 discordants (33 LG1-only, 57 D1-only): $\chi^2_{cc} = 5.878$, $p = 0.0153$; exact binomial on discordants $p = 0.0149$.
- **Cross-LLM arm lifts.** qwen→Sonnet on D1: +5.39 pt, paired McNemar $\chi^2_{cc} = 7.481$, $p = 0.006$ (significant). qwen→Sonnet on LG1: +1.51 pt, $p = 0.54$ (not significant). The asymmetry is the measurement underlying §4.8.1’s structural-effect reading.

- **Reproduction note.** The qwen-variant scripts need Apertis API access to `claude-sonnet-4-6` (traversal, LG1 only) and `qwen3.6-plus` (answer, both arms). The Sonnet-variant scripts need Apertis API access to `claude-sonnet-4-6` for both traversal and answer. Wall clock: qwen variant ~110 min; Sonnet variant ~123 min (D1-S ~24 min + LG1-S ~99 min). Total M3DocVQA LLM spend: \$8.84 across both variants. No GPU. The M3DocVQA corpus (PDFs, docling layout JSONs, spaCy-extracted KGs, ColPali indices) is excluded from the supplementary artefacts due to size and licence constraints; the full pipeline is documented in the four execution scripts.

References

- Dallas Card, Peter Henderson, Urvashi Khandelwal, Robin Jia, Kyle Mahowald, and Dan Jurafsky. 2020. With little power comes great responsibility. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Jaemin Cho, Debanjan Mahata, Ozan Irsoy, Yujie He, and Mohit Bansal. 2024. M3docrag: Multi-modal retrieval is what you need for multi-page multi-document understanding. *Preprint*, arXiv:2411.04952.
- Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, and Jonathan Larson. 2024. From local to global: A graph rag approach to query-focused summarization. In *arXiv preprint arXiv:2404.16130*.
- Manuel Faysse, Hugues Sibille, Tony Wu, Bilel Omrani, Gautier Viaud, Céline Hudelot, and Pierre Colombo. 2024. Colpali: Efficient document retrieval with vision language models. In *International Conference on Learning Representations (ICLR) 2025*.
- Jinlan Fu, See-Kiong Ng, Zhengbao Jiang, and Pengfei Liu. 2023. GPTScore: Evaluate as you desire. *arXiv preprint arXiv:2302.04166*.
- Zhihao Guo and 1 others. 2024. LightRAG: Simple and fast retrieval-augmented generation. *arXiv preprint arXiv:2410.05779*.
- Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. 2020. Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps. In *Proceedings of the 28th International Conference on Computational Linguistics (COLING)*.
- Haoyu Huang, Yongfeng Huang, Junjie Yang, Zhenyu Pan, Yongqiang Chen, Kaili Ma, Hongzhi Chen, and James Cheng. 2025. Retrieval-augmented generation with hierarchical knowledge. *Preprint*, arXiv:2503.10150. Verified 2026-05-07 against arXiv API (D38 Phase 0); replaces erroneous Jiao2025/2507.05714 entry — that arXiv ID belongs to a different "HiRAG" paper on instruction tuning.
- Omar Khattab and Matei Zaharia. 2020. Colbert: Efficient and effective passage search via contextualized late interaction over bert. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*.
- Aditya Kusupati, Gantavya Bhatt, Aniket Rege, Matthew Wallingford, Aditya Sinha, Vivek Ramanujan, William Howard-Snyder, Kaifeng Chen, Sham Kakade, Prateek Jain, and Ali Farhadi. 2022. Matryoshka representation learning. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*.
- Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. 2023. G-eval: Nlg evaluation using gpt-4 with better human alignment. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Quinn McNemar. 1947. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157.

- Rodrigo Nogueira, Zhiying Jiang, Ronak Pradeep, and Jimmy Lin. 2020. Document ranking with a pretrained sequence-to-sequence model. In *Findings of the Association for Computational Linguistics: EMNLP 2020*.
- Jiho Park and 1 others. 2025. Do chop-qa: Towards multi-hop reasoning over multimodal document collections. *Preprint*, arXiv:2508.15851. ArXiv ID carried forward from v3; manual verification recommended before camera-ready submission.
- Stephen Robertson and Hugo Zaragoza. 2009. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389.
- Yixuan Tang and Yi Yang. 2024. MultiHop-RAG: Benchmarking retrieval-augmented generation for multi-hop queries. *arXiv preprint arXiv:2401.15391*.
- Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2022. MuSiQue: Multihop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554.
- Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2023. Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL 2023)*.
- Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. 2018. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.